

Probability Theory and Statistics

Exercise 7 solutions

10.20. – 10.24.

Normal Distribution, Central Limit Theorem

1. Let X denote the waiting time (in hours) until the first shooting star is seen. By the problem text this is continuous and memoryless, hence exponential (the only memoryless continuous distribution). Let λ be its parameter, i.e. $X \sim \text{Exp}(\lambda)$. Seeing a shooting star within the first 20 minutes means $X < \frac{1}{3}$. Since for $x > 0$ the cdf of $\text{Exp}(\lambda)$ is $F_X(x) = 1 - e^{-\lambda x}$, the equality

$$1 - e^{-2/3} = F_X\left(\frac{1}{3}\right) = 1 - e^{-\lambda \cdot \frac{1}{3}}$$

gives $\lambda = 2$.

The first question asks for $\mathbb{P}(X < 1) = F_X(1) = 1 - e^{-\lambda \cdot 1} = 1 - e^{-2} \approx 0.8647$.

Because the waiting time is memoryless, the remaining time has the same distribution and variance as the original waiting time. Thus we need $\text{Var}(X)$. For $\text{Exp}(\lambda)$ we have $\text{Var}(X) = \frac{1}{\lambda^2}$, hence

$$\mathbb{D}^2(X) = \frac{1}{\lambda^2} = \frac{1}{2^2} = \frac{1}{4}.$$

2. Consider a washing machine of the given type, and let X be the time to first failure. By the text X is continuous and memoryless, hence exponential. Let $X \sim \text{Exp}(\lambda)$. We also know $\mathbb{E}(X) = 2$, hence $\lambda = \frac{1}{2}$ (since $\mathbb{E}(X) = 1/\lambda$ for an exponential).

The required probability is

$$\mathbb{P}(X > 3 \mid X > 2) = \mathbb{P}(X > 1) = \mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X < 1) = 1 - F_X(1) = e^{-\lambda} = e^{-1/2} \approx 0.6065.$$

3. Let $t = 0$ be 4 p.m. (when Dominik starts the program) and let X be the time (in hours) from then until the next crash. Then $X \sim \text{Exp}(\lambda)$ with $\lambda = 1/10$. From 4 p.m. to 8 a.m. is 16 hours, and from there to noon is 4 more hours, so we seek

$$\mathbb{P}(\text{no crash by 12} \mid \text{no crash by 8}) = \mathbb{P}(X > 20 \mid X > 16).$$

By memorylessness this equals $\mathbb{P}(X > 4)$, and by continuity

$$\mathbb{P}(X > 4) = \mathbb{P}(X \geq 4) = 1 - F_X(4) = e^{-\lambda \cdot 4} = e^{-0.4} \approx 0.6703.$$

Remark. Using $>$ vs. $\geq$ does not affect the result for a continuous law.

4. Let the parameters be λ_x and λ_y . Since $\mathbb{E}(\text{Exp}(\lambda)) = 1/\lambda$, the condition $\mathbb{E}(X) = 2\mathbb{E}(Y)$ gives $\frac{1}{\lambda_x} = \frac{2}{\lambda_y}$, i.e. $\lambda_y = 2\lambda_x$. Also $\mathbb{P}(Y < 1) = F_Y(1) = 1 - e^{-\lambda_y}$, while $\mathbb{P}(X > 1) = e^{-\lambda_x}$. The relation

$$3\mathbb{P}(X > 1) = 2\mathbb{P}(Y < 1) \iff 3e^{-\lambda_x} = 2(1 - e^{-\lambda_y})$$

together with $\lambda_y = 2\lambda_x$ gives

$$3e^{-\lambda_x} = 2(1 - e^{-2\lambda_x}).$$

Let $a = e^{-\lambda_x}$, so $e^{-2\lambda_x} = a^2$. Then $3a = 2(1 - a^2)$, i.e. $2a^2 + 3a - 2 = 0$. Hence

$$a = \frac{-3 + \sqrt{9 + 16}}{4} = \frac{-3 + 5}{4} = \frac{1}{2} \quad (\text{discard } a = -2 < 0).$$

Thus $e^{-\lambda_x} = \frac{1}{2}$, so $\lambda_x = \ln 2$, and therefore

$$\mathbb{E}(X) = \frac{1}{\lambda_x} = \frac{1}{\ln 2} \approx 1.4427.$$

5. *Solution.* (1 pt) The waiting time X is exponential. (1 pt) Indeed, the only continuous memoryless law is exponential. (1 pt) We are given $e^{-2} = \mathbb{P}(X \geq 1)$ (1 pt) $= 1 - \mathbb{P}(X < 1)$. (1 pt) Hence $F_X(1) = \mathbb{P}(X < 1) = 1 - e^{-2 \cdot 1}$. (1 pt) So the parameter is $\lambda = 2$ (e.g. by the standard table).

Solution 1. (1 pt) Let Y be the number (among the three friends) who wait at least half an hour. (1+1+1 pts) Then $Y \sim \text{Bin}(n = 3, p = \mathbb{P}(X \geq 1/2))$, (2 pts) where $\mathbb{P}(X \geq 1/2) = 1 - \mathbb{P}(X < 1/2) = e^{-1}$. (2 pts) We want $\mathbb{P}(Y \geq 2)$ (1 pt) $= \mathbb{P}(Y = 2) + \mathbb{P}(Y = 3)$ (1+1 pts) $= \binom{3}{2}(e^{-1})^2(1 - e^{-1}) + \binom{3}{3}(e^{-1})^3$ (1+1 pts) $= 0.2566 + 0.0498$ (1 pt) $= 0.3064$.

Solution 2. (0 pt) Let A, B, C be the events that Adam, Betty, and Cecilia wait at least half an hour. (1+1 pts) We need $\mathbb{P}$ that at least two of A, B, C occur, i.e. $\mathbb{P}((A \cap B) \cup (A \cap C) \cup (B \cap C))$. (2 pts) By inclusion-exclusion,

$$\mathbb{P}((A \cap B) \cup (A \cap C) \cup (B \cap C)) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap C) + \mathbb{P}(B \cap C) - 2\mathbb{P}(A \cap B \cap C).$$

(2 pts) This equals $3\mathbb{P}(A)^2 - \mathbb{P}(A)^3$, (1 pt) since the waiting times (hence A, B, C) are jointly independent (1 pt) and identically distributed. (1 pt) $\mathbb{P}(A) = \mathbb{P}(X \geq 1/2)$ (1 pt) $= 1 - \mathbb{P}(X < 1/2)$ (1 pt) $= e^{-1}$, (1 pt) hence the desired probability is $3(e^{-1})^2 - 2(e^{-1})^3 \approx 0.3064$.

6. Since Y is a nonnegative r.v. with finite mean (its range is finite), the identity $\mathbb{E}(Y) = \int_0^\infty (1 - F_Y(y)) dy$ applies. We can also compute both sides directly.

Mapping outcomes: $X = 1, 5 \Rightarrow Y = 2$; $X = 2, 4 \Rightarrow Y = 1$; $X = 3 \Rightarrow Y = 0$; $X = 6 \Rightarrow Y = 3$. With $\mathbb{P}(X = k) = 1/6$,

$$\mathbb{P}(Y = 0) = \mathbb{P}(Y = 3) = \frac{1}{6}, \quad \mathbb{P}(Y = 1) = \mathbb{P}(Y = 2) = \frac{1}{3}.$$

Hence $\mathbb{E}(Y) = 1 \cdot \frac{1}{3} + 2 \cdot \frac{1}{3} + 3 \cdot \frac{1}{6} = 1.5$. The cdf $F_Y(y) = \mathbb{P}(Y < y) = \sum_{k \in \text{Ran}(Y): k < y} \mathbb{P}(Y = k)$ is

$$F_Y(y) = \begin{cases} 0, & y \leq 0, \\ \frac{1}{6}, & 0 < y \leq 1, \\ \frac{3}{6}, & 1 < y \leq 2, \\ \frac{5}{6}, & 2 < y \leq 3, \\ 1, & y > 3. \end{cases}$$

Then

$$\int_0^\infty (1 - F_Y(y)) dy = \int_0^1 \left(1 - \frac{1}{6}\right) dy + \int_1^2 \left(1 - \frac{3}{6}\right) dy + \int_2^3 \left(1 - \frac{5}{6}\right) dy = \frac{5}{6} + \frac{1}{2} + \frac{1}{6} = 1.5.$$

7. Choose a so that f_X is a density: nonnegative and integrating to 1. As $f_X(x) = 2x$ on $(0, a)$ and 0 elsewhere,

$$1 = \int_{-\infty}^\infty f_X(x) dx = \int_0^a 2x dx = [x^2]_0^a = a^2,$$

hence $a = 1$. Then

$$\mathbb{E}(X) = \int_{-\infty}^\infty x f_X(x) dx = \int_0^1 2x^2 dx = \left[\frac{2}{3}x^3\right]_0^1 = \frac{2}{3}.$$

8. Let K be the tank capacity. We want $\mathbb{P}(X > K) < 0.05$. Since f_X vanishes outside $(0, 1)$,

$$\mathbb{P}(X > K) = \int_K^\infty f_X(x) dx = \int_K^1 5(1-x)^4 dx = \left[-(1-x)^5\right]_K^1 = (1-K)^5.$$

Thus $(1-K)^5 < 0.05$, i.e. $K > 1 - \sqrt[5]{0.05} \approx 0.45072$.

So the capacity should be at least 45,072 L.

For the mean weekly consumption,

$$\mathbb{E}(X) = \int_0^1 x \cdot 5(1-x)^4 dx.$$

Let $y = 1 - x$ ($dx = -dy$). Then

$$\mathbb{E}(X) = \int_0^1 (1-y) \cdot 5y^4 dy = \int_0^1 5(y^4 - y^5) dy = \left[y^5 - \frac{5}{6}y^6\right]_0^1 = \frac{1}{6} \approx 0.16667.$$

Thus the average weekly consumption is 16,667 L.

Remark. One can also expand $(1-x)^4$ via the binomial theorem and integrate termwise.

- a) For $x \in \mathbb{R}$, $F_X(x) = \int_{-\infty}^x f_X(t) dt$. If $x \leq 0$, then $F_X(x) = 0$; if $x > 1$, then $F_X(x) = 1$. For $0 < x \leq 1$,

$$F_X(x) = \int_0^x \frac{1}{2\sqrt{t}} dt = [\sqrt{t}]_0^x = \sqrt{x}.$$

In summary,

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ \sqrt{x}, & 0 < x \leq 1, \\ 1, & x > 1. \end{cases}$$

- b) Since $Y = X\sqrt{X} = X^{3/2}$ and $X \in (0, 1)$, we have $\text{Ran}(Y) = (0, 1)$. Hence $F_Y(y) = 0$ for $y \leq 0$ and $F_Y(y) = 1$ for $y > 1$. For $0 < y \leq 1$,

$$F_Y(y) = \mathbb{P}(X^{3/2} < y) = \mathbb{P}(X < y^{2/3}) = F_X(y^{2/3}) = \sqrt{y^{2/3}} = y^{1/3}.$$

Thus

$$F_Y(y) = \begin{cases} 0, & y \leq 0, \\ y^{1/3}, & 0 < y \leq 1, \\ 1, & y > 1. \end{cases}$$

c) Differentiating,

$$f_Y(y) = \begin{cases} 0, & y \leq 0, \\ \frac{1}{3y^{2/3}}, & 0 < y < 1, \\ 0, & y \geq 1. \end{cases}$$

d) $\mathbb{E}(Y) = \int_0^1 y \cdot \frac{1}{3}y^{-2/3} dy = \int_0^1 \frac{1}{3}y^{1/3} dy = \left[\frac{1}{4}y^{4/3} \right]_0^1 = \frac{1}{4}.$

e) Using the transformation formula directly from X ,

$$\mathbb{E}(Y) = \mathbb{E}(X\sqrt{X}) = \int_0^1 x\sqrt{x} \cdot \frac{1}{2\sqrt{x}} dx = \int_0^1 \frac{x}{2} dx = \left[\frac{x^2}{4} \right]_0^1 = \frac{1}{4}.$$

(As expected, this agrees with part d), and here it is simpler.)

9. First, get the cdf of $Y = X^2$; then differentiate. Clearly $Y \geq 0$, so for $y \leq 0$ we have $F_Y(y) = 0$. For $y > 0$,

$$F_Y(y) = \mathbb{P}(X^2 < y) = \mathbb{P}(X < \sqrt{y}) = F_X(\sqrt{y}) = 1 - e^{-\lambda\sqrt{y}},$$

since $X \geq 0$ for an exponential. Differentiating (chain rule),

$$f_Y(y) = F'_Y(y) = \frac{\lambda}{2\sqrt{y}} e^{-\lambda\sqrt{y}}, \quad y > 0,$$

and $f_Y(y) = 0$ otherwise.

For the mean, note $\mathbb{E}(Y) = \mathbb{E}(X^2)$. Using $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}X)^2$ and, for exponential, $\mathbb{E}X = 1/\lambda$, $\text{Var}(X) = 1/\lambda^2$, we get

$$\mathbb{E}(Y) = \mathbb{E}(X^2) = \text{Var}(X) + (\mathbb{E}X)^2 = \frac{1}{\lambda^2} + \frac{1}{\lambda^2} = \frac{2}{\lambda^2}.$$

10. Proceed as before: get cdfs, then densities. For $X \sim U(0, 1)$, $F_X(x) = x$ on $(0, 1)$ and equals 0 for $x \leq 0$, 1 for $x \geq 1$.

For $Y = \sqrt{2X}$ we have $\text{Ran}(Y) = (0, \sqrt{2})$. For $0 < x \leq \sqrt{2}$,

$$F_Y(x) = \mathbb{P}(\sqrt{2X} < x) = \mathbb{P}(X < x^2/2) = F_X(x^2/2) = x^2/2,$$

so

$$f_Y(x) = \begin{cases} x, & 0 < x < \sqrt{2}, \\ 0, & \text{otherwise.} \end{cases}$$

For $V = \ln \frac{1}{X}$ we have $\text{Ran}(V) = (0, \infty)$. For $x > 0$,

$$F_V(x) = \mathbb{P}\left(\ln \frac{1}{X} < x\right) = \mathbb{P}\left(\frac{1}{X} < e^x\right) = \mathbb{P}(X > e^{-x}) = 1 - \mathbb{P}(X \leq e^{-x}) = 1 - e^{-x},$$

hence

$$f_V(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & \text{otherwise,} \end{cases}$$

i.e. $V \sim \text{Exp}(1)$.

For $Z = \arctan(X)$ we have $\text{Ran}(Z) = (0, \pi/4)$. For $0 < x < \pi/4$,

$$F_Z(x) = \mathbb{P}(\arctan(X) < x) = \mathbb{P}(X < \tan x) = \tan x,$$

so

$$f_Z(x) = \begin{cases} \sec^2 x = \frac{1}{\cos^2 x}, & 0 < x < \pi/4, \\ 0, & \text{otherwise.} \end{cases}$$

11. Since

$$X \frac{\text{miles}}{\text{gallon}} = aX \frac{\text{km}}{\text{gallon}} = \frac{aX}{b} \frac{\text{km}}{\text{liter}} = \frac{100aX}{b} \frac{\text{km}}{100 \text{ km}} \frac{\text{liter}}{\text{liter}}$$

the liters/100 km consumption is $Y = \frac{100b}{aX}$. For $x > 0$,

$$F_Y(x) = \mathbb{P}\left(\frac{100b}{aX} < x\right) = \mathbb{P}\left(X > \frac{100b}{ax}\right) = 1 - F_X\left(\frac{100b}{ax}\right),$$

and $F_Y(x) = 0$ for $x \leq 0$. Differentiating (where applicable),

$$f_Y(x) = \frac{100b}{a x^2} f_X\left(\frac{100b}{ax}\right) \quad (x > 0), \quad f_Y(x) = 0 \text{ otherwise.}$$

Remark. Even if F_X is not differentiable everywhere, the displayed f_Y still integrates to F_Y by substitution $u = \frac{100b}{ax}$, so it correctly represents the density. Cdfs of continuous r.v.s are differentiable almost everywhere, so $f_Y = F'_Y$ holds a.e.

12. (a) $f_X(x) = 0$ outside $(1, 2)$. On $(1, 2)$ the density is the derivative of the cdf. For $x \in (1, 2)$, $f_X(x) = F'_X(x) = \frac{2}{3}x$.

$$(b) \mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x) dx = \int_1^2 x \cdot \frac{2}{3}x dx = \int_1^2 \frac{2}{3}x^2 dx = \left[\frac{2}{9}x^3\right]_1^2 = \frac{16}{9} - \frac{2}{9} = \frac{14}{9}.$$

(c) Idea: use the formula for transformed expectations in the continuous case. Let $k \geq 1$. Then $\frac{3}{2}\mathbb{E}(X^k) = \frac{3}{2} \int_{-\infty}^{\infty} x^k f_X(x) dx = \int_1^2 x^{k+1} dx$. For $Y \sim U(1, 2)$, $f_Y(x) = 1$ on $(1, 2)$ and 0 otherwise, hence $\mathbb{E}(Y^{k+1}) = \int_1^2 x^{k+1} dx$. Therefore $\frac{3}{2}\mathbb{E}(X^k) = \mathbb{E}(Y^{k+1})$.

13. a) Since X is continuous, $\mathbb{P}(X > 3) = \mathbb{P}(X \geq 3) = 1 - \mathbb{P}(X < 3) = 1 - F_X(3)$. For $\text{Exp}(\lambda)$, $F_X(x) = 1 - e^{-\lambda x}$ ($x > 0$). Thus

$$e^{-6} = \mathbb{P}(X > 3) = 1 - (1 - e^{-\lambda \cdot 3}) = e^{-3\lambda},$$

so $\lambda = 2$.

b) $\mathbb{P}(X < 2) = F_X(2) = 1 - e^{-\lambda \cdot 2} = 1 - e^{-4} \approx 0.9817$.

c) $\mathbb{E}(X) = 1/\lambda = 1/2$.

14. $X \sim U(0, 16.4)$, hence

$$\mathbb{D}(X) = \frac{b-a}{2\sqrt{3}} = \frac{16.4}{2\sqrt{3}} \approx 4.7343.$$

Since the law depends only on line length, the answer is the same for any line of length 16.4 km.

15. Using the density of X derived in Problem Set 5, Problem 15,

$$\begin{aligned}\mathbb{E}(X) &= \int_1^2 x \left(\frac{1}{\sqrt{x-1}} - 1 \right) dx = \int_1^2 \left(\sqrt{x-1} + \frac{1}{\sqrt{x-1}} - x \right) dx = \\ &= \left[\frac{2}{3}(x-1)^{3/2} + 2\sqrt{x-1} - \frac{x^2}{2} \right]_1^2 = \frac{7}{6}.\end{aligned}$$

16. Using the density of Y from Problem Set 5, Problem 16,

$$\mathbb{E}(Y) = \int_0^1 x(6x - 6x^2) dx = 6 \int_0^1 (x^2 - x^3) dx = 6 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = 6 \cdot \frac{1}{12} = \frac{1}{2}.$$

17. a) Note $Y = \max\{0, X\}$ equals 0 iff $X \leq 0$, otherwise $Y = X > 0$. Thus for $x \leq 0$, $F_Y(x) = 0$, while for $x > 0$,

$$\begin{aligned}F_Y(x) &= \mathbb{P}(Y < x) = \mathbb{P}(Y = 0) + \mathbb{P}(0 < Y < x) = \mathbb{P}(X \leq 0) + \mathbb{P}(0 < X < x) \\ &= \mathbb{P}(X < x) = F_X(x).\end{aligned}$$

Hence

$$F_Y(x) = \begin{cases} 0, & x \leq 0, \\ F_X(x), & x > 0. \end{cases}$$

b) For $x \in \mathbb{R}$,

$$F_Z(x) = \mathbb{P}(Z < x) = \mathbb{P}(-X < x) = \mathbb{P}(X > -x) = 1 - \mathbb{P}(X \leq -x).$$

Note $\mathbb{P}(X \leq -x)$ is the right limit $F_X(-x+0)$, hence

$$F_Z(x) = 1 - F_X(-x+0).$$

(Indeed, for any t , $\mathbb{P}(X \leq t) = \lim_{y \downarrow t} F_X(y) = F_X(t+0)$ by monotone continuity from above.)

c) Since $V = |X| \geq 0$, $F_V(x) = 0$ for $x \leq 0$. For $x > 0$,

$$\mathbb{P}(V < x) = \mathbb{P}(|X| < x) = \mathbb{P}(-x < X < x) = \mathbb{P}(X < x) - \mathbb{P}(X \leq -x) = F_X(x) - F_X(-x+0).$$

Hence

$$F_V(x) = \begin{cases} 0, & x \leq 0, \\ F_X(x) - F_X(-x+0), & x > 0. \end{cases}$$

d) $W = \min\{0, -X\} = -\max\{0, X\} = -Y$. Using parts (a) and (b),

$$F_W(x) = \mathbb{P}(-Y < x) = 1 - F_Y(-x+0) = \begin{cases} 1 - F_X(-x+0), & x \leq 0, \\ 1, & x > 0. \end{cases}$$